

Science 24

UNIT A WORKBOOK



SENIOR HIGH SCHOOL

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Chapter 1: Useful Chemistry

Define the following:

Synthetic

Pesticides

Herbicides

Insecticides

Synthetic Fibers

Synthetic Polymers

Alloy

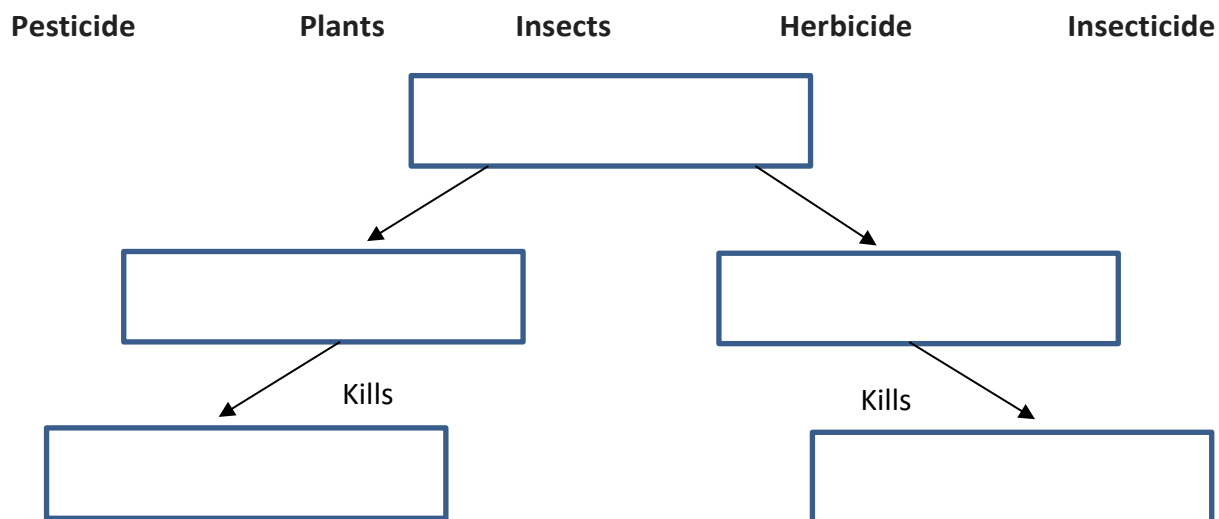
HHPS

WHIMIS

MSDS

1.1 ***Chemistry Then and Now***

1. Complete the chart using the following 5 words (pg. 6)



2. What does synthetic mean?(pg.6)

3. How do chemists help modern farmers?(pg. 6)

4. How does modern soap differ from soap made 100 years ago?(pg. 7)

5. Name 2 synthetic and 2 natural fibers. Describe at least two advantages of synthetic fibers. (pg. 8)

6. What are synthetic polymers? (pg. 9)

7. Give two examples of natural polymers and 2 examples of synthetic ones.(pg. 9)

8. Synthetic polymers are used in many products. Name three synthetic polymers and briefly explain how they have changed daily life. (pg. 9)

9. In winter you choose clothes with fabrics that will keep you warm. (pg. 9)

a. What materials keep you warm today?

b. What materials would you have used 100 years ago? (pg. 8)

10. What metals are found in stainless steel utensils? How would your life be

different if kitchen utensils were not made from stainless steel? (pg. 10-11)

11. Many household cleaners rely on chemical changes to do their work. What chemical change occurs when you add drain cleaner to a clogged pipe? (pg. 12)

12. Yeast helps bread to rise. Explain the chemical change that yeast produces and how this makes bread rise. (pg. 13)

13. Soap cleans your hair but why won't it make your hair smooth and shiny? (pg. 16)

14. Choose two different workplaces and identify five hazardous substances you would find in each one.

15. Why is it important for workers at a restaurant to understand WHMIS? (pg. 19)

Chapter 2: Common Reactions to Our Lives

Define the Following:

Chemical reaction

Reactants

Products

Combustion

Neutralization

Precipitate

Energy changes

Exothermic reaction

2.1 Useful Reactions

1. List the steps that happen when a banana ripens. (pg. 24)
2. What is responsible for the color change when using bleach? (pg. 24)
3. Sunlight can break down sodium hypochlorite (bleach). Observe the plastic containers used for bleach. Why will you never see a transparent bleach container?
4. How is baking bread a chemical reaction? (pg. 25)
5. What safety equipment should you use when handling chlorine in a pool? (pg. 25)

6. You have a glowing splint, and it reignites in a test tube. What gas is in the test tube?
7. Label the reactants and products, and state whether they are a neutralization or a combustion reaction in the following equations. (pg. 29)
- a. Magnesium hydroxide and Hydrochloric acid combine to form water and magnesium chloride.
- b. Methane and oxygen combine to form carbon dioxide and water.
8. Give three examples of combustion reactions that you encounter in a day. List as many products and reactants that you can. (pg. 29)
9. Fishy odors on your hands are caused by a base and can be eliminated with lemon juice. Is this an example of a combustion or a neutralization reaction? Why? What are the reactants? (pg. 29)

10. List 5 things you could observe to tell if there was a chemical change. (pg. 34)

11. In the following scenarios state whether a chemical change occurred and state why you know it did. Use the reasons from the previous question.

- a. Hydrochloric acid is added to Magnesium and Hydrogen gas bubbles out.
- b. Two liquids are mixed together and a solid forms at the bottom of the test tube
- c. Phenolphthalein, a colorless indicator, changes color in the presence of an acid
- d. Water freezes and forms ice
- e. Iron is heated until it becomes molten
- f. Two chemicals are mixed together, and you can smell spearmint
- g. Two liquids are mixed together and ice forms on the outside of the test tube

12. How do exothermic and endothermic reactions differ? How are they similar?
(pg. 35)

13. Some exothermic reactions can be dangerous. Give the reason behind this statement.
(pg. 35)

14. For the following situations determine if the reaction is exothermic or endothermic.

a. Methane is burnt.

b. When mixing two gases a loud explosion is heard.

c. Sodium acetate reacts and creates ice.

d. A plant absorbs energy from the sun.

e. Glucose in your cell produces energy.

Chapter 3: Types of Chemical Reactions

Define the following:

Word Equation

Simple Composition

Chemical Name

Chemical Formula

Common Name

Law of Conservation of Mass

Balanced Chemical Equation

Coefficients

3.1 Word Equations

1. For each word equation write what you would say when reading it. (pgs. 42-43)

a. Solid zinc + hydrochloric acid $\rightarrow$ hydrogen gas + zinc chloride

b. Hydrochloric acid + sodium acetate $\rightarrow$ acetic acid + sodium chloride

2. For each of the following reactions identify whether it is a simple composition reaction, a simple decomposition reaction, a combustion reaction, or a neutralization reaction.
(pgs. 44 - 49)

a. Methane + oxygen $\rightarrow$ carbon dioxide + water + energy

b. Hydrogen gas + oxygen gas $\rightarrow$ water

c. Water $\rightarrow$ hydrogen gas + oxygen gas

d. Hydrochloric acid + sodium hydroxide $\rightarrow$ salt and water

e. $\text{NaCl} \rightarrow \text{Na} + \text{Cl}$

f. $\text{C}_2\text{H}_5\text{OH} + \text{O}_2 \rightarrow \text{H}_2\text{O} + \text{CO}_2$

g. Solid sodium hydroxide + liquid acetic acid $\rightarrow$ sodium acetate (a salt) + water

3. Draw a diagram representing a composition reaction. (pg. 45)

4. Draw a diagram representing a decomposition reaction. (pg. 45)

5. Why do scientists use chemical formulas? (pg. 50)

6. If something is dissolved in water what subscript should you use? (pg. 50)

7. What is the common name for the following compounds?(pg. 51)

a. NaCl

b. $\text{CaCO}_{3(s)}$

c. $\text{C}_2\text{H}_5\text{OH}_{(l)}$

d. $\text{C}_6\text{H}_{12}\text{O}_{6(s)}$

e. $\text{H}_2\text{O}_{(l)}$

f. $\text{NaHCO}_{3(s)}$

8. If I have 10g of reactants how many grams of product should I have?
Why? (pg.52)

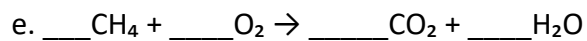
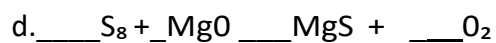
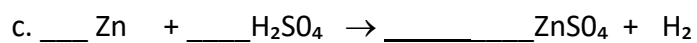
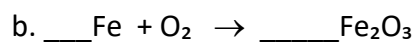
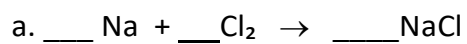
9. The following reaction takes place: (pg. 52)



a. What would the mass of the reactants be if there was 50g of product?

b. If there was 40g of HCl how much NaOH was there?

10. Balance the following equations by filling in the correct coefficient:



Chapter 4: Reactions and the Environment

Define the following:

Greenhouse gases

Global warming

Climate change

Pollutants

Acid deposition

Sulfur dioxide

Nitrous oxide

Liming

Scrubber

Corrosion

Rusting

Galvanization

Sacrificial metal

4.1 Fossil Fuel Combustion

1. What is the reaction we use to convert fossil fuels into energy? pg. 60
2. What are the products of combustion? pg. 60
3. How does combustion differ from Incomplete combustion? pg. 60 and 61
4. Name 4 products of incomplete combustion. pg. 61
5. What causes incomplete combustion to occur? pg. 61

6. Why can incomplete combustion be deadly? pg 61

7. List 3 natural greenhouse gases. pg. 62

8. Describe three negative effects of global climate change. pg. 62

9. What are the major contributors to acid deposition? pg. 64

10. Name two gas emissions that are linked to acid deposition. pg. 65

11. What is the acid in the stomach that is neutralized? pg. 66

12. What neutralizes chemical neutralizes the acid in your stomach? pg. 66

13. What are the products of a neutralization reaction? pg. 66

14. What is liming and why do we do it? pg. 66

15. What is the P.H. range of an acid? pg. 66

16. How can we reduce the amount of acid deposition? pg. 67

17. What are scrubbers? What do they do? pg. 68

18. Describe how driving a car contributes to the problem of acid precipitation. pg. 67

19. Why is burning coal causing acid deposition? pg. 68

20. Why is the corrosion of iron and steel a serious problem? pg. 69

21. What type of reaction is rusting? pg. 69

22. List some properties of Iron III oxide (Rust). pg. 69

23. Why does metal rust even after its outer layers have corroded? pg. 69

24. List 3 ways we can prevent rusting. pgs. 72-73

25. Why does galvanization work? pg. 72